Extra Vaccines

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- Rotavirus vaccine
- Pneumococcal vaccine
- Varicella vaccine
- Hepatitis A vaccine
- Meningococcal vaccine
- Influenza vaccine

اسم التطعيم	النوع	الجرعة	عمر الطفل
الروتا **Rotavirus vaccine**	عن طريق الفم	الاولى	شهرين
		الثانية	4 شهور
		التالتة	6 شهور
المكورات الرئوية **PCV13**	حقن / عضل	الاولى	شهرين
		التانية	4 شهور
		التالتة	6 شهور
		المنشطة	15-12 شهر
الجديري المائي **Varicella vaccine**	حقن / تحت الجلد	الاولى	15 -12 شهر
		التانية	3 اشهر على الاقل بعد الجرعة الاولى
الكبدي الوبائي أ **HAV**	حقن / عضل	الاولى	23-12 شهر
		التانية	12-6 شهر بعد الجرعة الاولى
الحمى الشوكية **Meningococcal polysaccharide vaccine**	حقن / عضل	الاولى	سنتان
		التانية	5- 4 سنوات
الانفلونزا الموسمية Seasonal influenza	حقن/عضل	سنويا خلال فصل الخريف	

Rotavirus vaccine

Rotavirus vaccination

(Minimum age : 6 weeks)

(Minimum interval between doses : 4 weeks)

➨ Routine vaccination

-Rotarix : 2-dose series at age 2 and 4 months

-Rotateq : 3-dose series at age 2, 4, 6 months

-If any dose in the series is either **Rotateq** or unknown,

Default to 3-dose series. (**they are interchangeable**)

Catch-up vaccination :

- Do not start the series on or after age 15 weeks , 0 days.
- The maximum age for the final dose is 8 months , 0 days.

According to the package inserts the maximum age for last dose of Rotateq is 32 weeks & the maximum age for Rotarix is 24 weeks. According to ACIP recommendations the maximum age for last dose of rotavirus vaccine is 8 months , 0 days. Eight months 0 days is older than age 24 weeks and may be older than age 32 weeks. Should I follow the package labels or the ACIP recommendations ?

- ACIP recommendations and package inserts do not always match.
- Occasionally, ACIP may use different data to formulate its recommendation, or try to add flexibility to its recommendation, which results in a recommendation different than the package insert.
- Published recommendations of national advisory groups (such as ACIP or AAP's committee on infectious diseases) should be considered equally as authoritative as those on the package insert. You should consider 8 months 0 days as the maximum age for a dose of rotavirus vaccine.

Precautions of Rotavirus vaccine administration :

- **Co-administration** of the oral polio and rotavirus vaccines may be accompanied by an interference on immunogenicity.
- A minimum interval of **2 weeks** is recommended for the administration of rotavirus vaccination and oral polio vaccine.
- Early clinical trials suggested that oral poliovirus vaccine , when **administered concomitantly** with rotavirus vaccine , reduces rotavirus seroconversion rates **after 1st rotavirus dose** with modest or **non-significant interference after completion of the full rotavirus vaccine course**.
- If an incomplete dose of an oral vaccine is administered for any reason (e.g , infant spits or regurgitates the vaccine) a replacement dose should not be administered. " Pinkbook : Vaccine Administration – CDC " , "**Canadian Immunization Guide**"

Pneumococcal vaccine

Pneumococcal vaccination (minimum age : 6 weeks) : PCV 13

Prevenar 13 , Synflorix (10 serotypes)

Routine vaccination with PCV :

4-dose series at 2 , 4 , 6 , 12-15 months

Catch – up vaccination with PCV:

- If the series has been started at 6 months of age or older and younger than one year : 3 dose series ,preferably 8 weeks apart (minimum 4 week interval) , then the last dose between the ages of 12 through 15 months (given at least 8 weeks after the second dose.
- If the series has been started at one year of age or older and younger than 2 years : 2 dose series (8 weeks apart).
- Healthy children age 24-59 months with any incomplete PCV series : 1 dose PCV (patients with immunocompromising or chronic medical conditions : 2 doses of PCV at least 8 weeks apart , give one dose of PPSV23 at least 8 weeks after last dose of PCV)

- N.B : PneuomoRecs VaxAdvisor application for pneumococcal vaccine Catch-up guidance (available for android and IOS users , approved by CDC).

Can we consider PREVENAR 13 and SYNFLORIX interchangeable ?

- **Yes** , The Theoretical serotype coverage of SYNFLORIX is similar to that of Prevenar.
- **Pan American Health Organization (PAHO)** ; Available evidence to date indicates significant impact of both PCV10 & PCV13 in the outcomes studied , with no evidence of the superiority of one vaccine over the other on pneumonia , IPD or meningitis hospitalization reduction in children under 5 years old.

Precautions of pneuomococcal vaccine administration :

- Giving **inactivated influenza** vaccine with **PCV13** increases **the risk for febrile convulsions in children** according to a vaccine safety Datalink study
- **Concomitant administration** is not preferred , They should be separated by **any interval (i.e,1 - 2 weeks)** to **decrease the risk of febrile convulsions in children.**

Varicella vaccine

Varicella vaccination
(minimum age : 12 months)
Routine vaccination
2 dose series : Varivax or Varilrix
They are interchangeable

Dosage		
Age	**Children** **12 months to 12 years**	**Adolescents and Adults** **13 years and older**
1st Dose	**0.5 – ml dose administered subcutaneously (or IM)**	**0.5 -ml dose administered subcutaneously on elected date (or IM)**
2nd Dose	**0.5 – ml dose , a minimum interval of 3 months after the first dose** **(2 months " grace period ")**	**0.5 – ml dose to be administered , a minimum interval of 4 weeks after the first dose**

Varicella Post Exposure prophylaxis:

- The Advisory Committee on Immunization Practices **(ACIP)** recommends that after being exposed to varicella or herps zoster , people **who do not have evidence of immunity** and are **eligible for vaccination should get varicella vaccine :**

- Ideally, the vaccine should be given within **3-5 days** after the person is exposed. This may prevent varicella or make it less severe.
- Even if it has been more than 5 days, the vaccine should still be offered. This will provide protection against varicella if a person is exposed again in the future.

Other types of post-exposure prophylaxis : UK Health Security Agency - Antivirals (aciclovir or valacicolvir)

- **Oral aciclovir (or valaciclovir) is now the first choice of PEP for susceptible immunosuppressed individuals, all pregnant women at any age of pregnancy and infants at high risk. Oral acyclovir (or valaciclovir) should be given from day 7 to day 14 after exposure . The day of exposure is defined as the date of onset of the rash if the index is a household contact and date of first or only contact if the exposure is on multiple or single occasions respectively.**

- **If the patients presents after day 7 of exposure , a 7-day course of antivirals can be started up to day 14 after exposure , if necessary.**
- **In a study evaluating the comparative effectiveness of a 7-day course of aciclovir given either immediately after exposure or starting at day 7 after exposure to healthy children , the incidence and severity of varicella infection was significantly higher in those given aciclovir immediately after exposure (10 of 13 ; 77%) who received aciclovir immediately developed clinical varicella compared with 3 of 14 (21%) who started aciclovir at day 7.**
- **A 7-day post exposure prophylaxis course of aciclovir or valacicolvir is therefore recommended to start from day 7 after exposure.**
- **PEP following second exposure :**
  If there is a second or subsequent exposure to chickenpox further courses of antivirals can be initiating starting 7 days after the date of onset or exposure

Recommended doses of oral antivirals

	Oral aciclovir	Oral Valaciclovir
Infants over 4 weeks to children under 2 years age	10 mg/kg 4 times daily , day 7 to 14 after exposure	Not recommended
Children 2 to 17 years of age	10 mg /kg (up to a maximum of 800 mg) , 4 times daily, from days 7 to 14 after exposure	20mg/kg (up to a maximum 1000 mg) 3 times daily , from days 7 to 14 after exposure
Adults	800mg 4 times daily , from days 7 to 14 after exposure	1000mg 3 times daily , from days 7 to 14 after exposure

individuals on long term acyclovir or valaciclovir prophylaxis , for example , post-haematopoietic stem cell transplant , may require their dose of aciclovir to be temporarily increased to the dosage as given in Table.

Human varicella- zoster immunoglobulin (VZIG)

- For individuals who are unable to take oral antivirals (individuals with renal impairment or intestinal malabsorption , for example inflammatory bowel disease, and for susceptible neonates exposed within one week of delivery (in utero or post-delivery)VZIG should be considered
- If VZIG is considered , it's important to demonstrate that the patient will benefit from the blood product by demonstrating that they are sero-negative with VZV **IgG antibody levels < 150 mIU / ml for immunosuppressed patients and < 100 mIU/ ml for pregnant women.**
- For immunosuppressed patients only , if time does not permit quantitative testing , a qualitative test must be performed and shown to be negative or equivocal. Similarly for pregnant women who are unable to take antivirals due to renal impairment , intestinal malabsorption or hyperemesis , if time does not permit quantitative testing , a qualitative test must be performed or shown to be negative.

Dosage and timing of VZIG for prophylaxis

0 to 5 years	250 mg	By slow intramuscular injection
6 to 10 years	500 mg	
11 to 14 years	750 mg	
15 years and older	1000 mg	

When a large-volume injection uch as VZIG is to be given , it should be administered deep into a large muscle mass. If more than 3 ml is to be given to young children and infants , or more than 5 ml to older children and adults , the immunoglobulin should be divided into smaller amounts and given into different sites . The upper outer quadrant of the buttock can be used to administer VZIG injection in neonates and infants.

VZIG should ideally be administered within 10 days (preferably 7 days for neonates and immunosuppressed contacts) of the day of exposure. The day of exposure is defined as the date of onset of the rash if the index is a household contact and date of first or only contact if the exposure is on multiple or single occasions respectively.

Further doses of VZIG following second exposure

- Individuals who have previously received VZIG or IVIG as VZV post-exposure prophylaxis require **a new risk assessment if a second exposure occurs :**

- Within 3 weeks of administration of VZIG or IVIG , further PEP is not required
- between 3 and 6 weeks following administration of VZIG or IVIG , further PEP (dose of VZIG) should be administered without further testing
- More than 6 weeks following administration of VZIG or IVIG , retesting of a new sample is required

Intravenous immunoglobulin (IVIG)

- Contacts with **bleeding disorders who cannot receive antivirals or be given an intramuscular injection** should given intravenous human normal immunoglobulin (IVIG) at a dose of 0.2 g per kg body weight (4 ml / kg for a 5 % solution) instead. This will produce serum VZV antibody levels equivalent to those achieved with VZIG.
- IVIG should ideally be administered within **10 days (preferably 7 days for neonates and immunosuppressed contacts)** , of the first contact, but can be given later if necessary.

Precautions of varicella vaccine administration :

- If 2 different live attenuated injectable vaccines are given < 28 days apart, the one given second should be repeated ≥ 28 days after the second or invalid dose. This would apply specifically to doses of **MMR** and **varicella**, if not administered on the same day.

What are the recommended intervals between administration of antibody-containing products and measles-or varicella-containing vaccine,by product and indication for vaccination ?

Product / indication	Dose (IgG/Kg) and route	Recommended interval before measles- or live varicella – containing vaccine administration
Blood transfusion – RBCs washed	10 ml/kg ,negligible IgG/Kg IV	None
Blood transfusion – RBCs , adenine –saline added	10 ml/kg (10 mg IgG /Kg) IV	3 months
Blood transfusion –Packed RBCs(hematocrit 65%)	10 ml/kg (60 mg IgG /Kg) IV	6 months
Blood transfusion – Whole blood (hematocrit35% - 50%)	10 ml/kg (80-100 mg IgG /Kg) IV	6 months
Blood transfusion-Plasma/platelet products	10 ml/kg (160 mg IgG /Kg) IV	7 months
Botulinum Immune Globulin Intravenous (Human)	1.0 ml/kg (50 mg IgG /Kg) IV	6 months
Cytomegalovirus IGIV	150 mg/kg maximum	6 months
Hepatitis A IG – Contact prophylaxis	0.1 ml/kg (16.5 mg IgG /Kg) IM	6 months
Hepatitis A IG –International travel < 1 month stay	0.1 ml/kg (16.5 mg IgG /Kg) IM	6 months
Hepatitis A IG – International travel ≥ 1 month stay	0.2 ml/kg (33 mg IgG /Kg) IM	6 months
Hepatitis B IG	0.06 ml/kg (10 mg IgG /Kg) IM	3 months
IGIV-replacement therapy for immune deficiencies	300-400 mg/kg IV	8 months
IGIV- Immune thrombocytopenic purpura treatment	400 mg/Kg IV	8 months
IGIV- Postexposure varicella prophylaxis	400 mg/Kg IV	8 months
IGIV-PE measles prophylaxis for immune compromised contacts	400 mg/Kg IV	8 months
IGIV- Immune thrombocytopenic purpura treatment	1000 mg/Kg IV	10 months
IGIV-Kawasaki disease	2 g/Kg IV	11 months
Measles prophylaxis IG-Standard (i.e nonimmunocompromised) contact	0.50 ml/kg (80 mg IgG/Kg) IM	6 months
Monoclonal antibody to RSV F protein (e.g Synagis [Medimmune])	15 mg/kg IM	None
Rabies IG	20 IU/Kg (22 mg IgG/Kg) IM	4 months
Tetanus IG	250 units (10 mg IgG/Kg) IM	3 months
Varicella IG	125 units /10 kg (60-200 mg IgG /Kg) IM ,maximum 625 units	5 months

According to Canadian guidelines :

- For optimum immune response to **MMR** or **Varicella containing vaccine**, the vaccine should be administered **at least 14 days** prior to administration of an Ig preparation or blood product , or the vaccine administration delayed until the antibodies in the Ig preparation or blood product have degraded(refer to table) . if the interval between the administration of any of these vaccines and subsequent administration of an Ig preparation or blood product is less than 14 days , or **if these vaccines are administered before the antibody has degraded , repeat the vaccine dose after the recommended interval.**

According to Canadian guidelines :

- There is minimal or no interaction between blood products or Ig preparations , and :
- Inactivated vaccines
- live oral vaccines (rotavirus , oral typhoid vaccines)
- live intranasal vaccine (live attenuated influenza vaccine)
- BCG vaccine
- Yellow fever vaccine

These vaccines may be given concomitantly with , or at any time before or after , an Ig preparation or blood product has been administered.

Hepatitis A vaccine

Hepatitis A vaccination :
(minimum age : 12 months for routine vaccination)

-The available products are Havrix 1440 , Havrix 720 Junior , Avaxim , and Avaxim Pediatric.

-These products are considered interchangeable at any scheduled dose for children or adults , using age-specific dosage for the particular product.

Routine vaccination:

-2 dose series (minimum interval : 6 months) at age 12- 23 months.

Catch up vaccination :

- Unvaccinated people through age 18 years should complete a 2 dose series (minimum interval : 6 months).
- People who previously received 1 dose at age 12 months or older should receive dose 2 at least 6 months after dose 1.
- Resume the series no matter when the first dose was received , you don't need to restart a vaccine series because an interval is longer than recommended. Doses given even years later than recommended are still valid because the body has "immunologic memory"

Special situation
(International travel)

- People travelling to or working in countries with high or intermediate endemic hepatitis A

- Infants age 6-11 months : 1 dose before departure ;

Revaccinate with 2 doses (separated by at least 6 months) between age 12- 23 months.

- Unvaccinated age 12 months or older : Administer dose 1 as soon as travel is considered.

Hepatitis A Post-Exposure Prophylaxis Guidance :

Postexposure prophylaxis (PEP)

All nonimmune people who are exposed to hepatitis A virus (HAV) and have not been previously infected or vaccinated should receive PEP within 14 days after the date of last exposure.

Definition of hepatitis A immunity:

People are considered immune if they have :

- received 2 doses of HAV vaccine; or

- a history of IgM or total anti-HAV positivity during or up to four months after clinically compatible illness ; or

-are igG anti-HAV positive

PEP recommendations

- People ≥ 12 months of age , regardless of pre-existing medical conditions , SHOULD receive a dose of single- antigen HAV vaccine. In addition to vaccine , a dose of intramuscular (IM) immune globulin (IG) 0.1ml/kg > 40 years based on the provider's risk assessment.
- People ≥ 12 months of age who have chronic liver disease and/or immunocompromised SHOULD receive a dose of intramuscular immune globulin (IMIG) (0.1ml/kg) and a dose of HAV vaccine.
- Infants < 12 months of age and/or people who are allergic to vaccine component) SHOULD receive intramuscular immune globulin (IMIG) (0.1ml/kg) instead of HAV vaccine.
- People receiving both vaccine and IG for post-exposure prophylaxis may receive them simultaneously , or they may receive whichever product is available first and the second product when it is available, providing it is administered within the 14-day PEP window.
- Vaccine and IG should be administered at anatomically distant sites (such as different limbs).

Meningococcal vaccine

Different Types of Meningococcal Vaccines :

➨ **Mencevax ACWY :**

- Meningococcal polysaccharide vaccine
- Neisseria meningitides serotypes A,C,W-135 and Y
- Not recommended in children < 2 years
- Repeated every 2-3 years in areas of high risk

N.B : Administered in schools in Egypt every 2-3 years

Menactra (Serogroup : ACWY-DT)

- Meningococcal polysaccharide Diphtheria Toxoid conjugate vaccine
- Neisseria meningitides serotypes A,C,W-135 and Y
- 9-23 months : 2 doses , 3 months apart
- 2-55 years : single dose
- A single booster dose may be given to individuals 15-55 years at continued risk for meningococcal disease.

Nimenrix : Serogroup : ACWY-TT

Age Group	Primary Immunisation	Booster
Infants from 6 weeks to less than 6 months of age	Two doses , each of 0.5 ml with the first dose given from 6 weeks of age , with an interval of 2 months between doses	At 12 months of age
Unvaccinated infants from 6 months to less than 12 months of age	One dose of 0.5 ml given from 6 months of age	At 12 months of age with a minimum interval of at least 2 months after the primary dose
Children from 12 months of age, adolescents and adults	One dose of 0.5 ml	Not routinely administered

Can a different brand of quadrivalent (A,C,W135,Y)meningococcal conjugate vaccines (4vMenCV) be used to complete a vaccination course (i.e are the brands interchangeable), particularly if there are shortages in vaccine supply ?

- **It's preferable to use the same brand of 4vMenCV when giving subsequent doses , especially for completing a primary vaccination course for young children. However , a different brand may be substituted in cases where the initial brand used is unavailable or not known. (National Centre For Immunization Research & Surveillance, NCIRS Fact sheet , Australia)**

Special consideration for use of Meningococcal conjugate vaccine and prevenar 13 :

- Where both **Prevenar 13** and **Menactra** are to be given , administration of Menactra must be **at least 4 weeks before or after** a dose of prevenar13.
- regarding use of **Menactra for people with functional or anatomic asplenia or HIV :**

- **Before the age of 2 years ;** children with functional or anatomic asplenia or HIV **shouldn't receive Menactra**. This timing avoids interference with immunologic response to the infant series of pneumococcal conjugate vaccine (PCV)
- **People 2 years of age or older** with functional or anatomic asplenia or HIV **shouldn't receive Menactra at the same time as PCV** , they should receive **Menactra 4 weeks after completion of all PCV doses**. **If these vaccines are inadvertently administered on the same day , PCV13 should be repeated a minimum of 8 weeks later.**

- Nimenrix may be co-administered with prevenar 13 **(The Australian immunization Handbook).**

Special considerations for concomitant use of Meningococcal conjugate vaccine and DTaP :

- Children can receive **Menactra (ACWY-DT)** or **Nimenrix (ACWY-TT)**before **(at least 4 weeks before)** or **concomitantly with DTaP vaccines**
- This timing avoids interference with the immunologic response to the meningococcal vaccine antigens that occurs when administering Menactra after DTaP.
- Even though ACIP recommends that Menactra should be given either **before(at least 4 weeks before) , or at the same visit , or at least 6 months after DTaP**, **there is no evidence to support repeating the dose of Menactra if administered at a time shorter than that recommended interval.**

N.B : The anti-diphtheria response was much higher after concomitant administration of Td with Menactra compared to the administration of Td followed by Menactra 28 days later.

(CDC , European Medicines Agency , Canadian immunization guide)

Special considerations for use of Meningococcal conjugate vaccine and polysaccharide meningococcal vaccine :

- Eligible individuals previously vaccinated with a polysaccharide meningococcal vaccine should be given meningococcal quadrivalent conjugate ; this should be offered at least 6 months after vaccination with polysaccharide meningococcal vaccine.

Influenza vaccine

Recommeded doses of influenza vaccine by age group : Influvac Tetra or Vaxigrip Tetra

Age	Dose	Number of doses needed in 1st year of influenza vaccination
6 months to < 9 years	0.5 ml	2 (given 4 weeks apart)
≥ 9years	0.5 ml	1
People of any age who have recently had haematopoietic stem cell transplant or solid organ transplant	0.5 ml	2 (given 4 weeks apart) in 1st year after transplant

Timing of vaccination

- For most people who need only one dose of influenza vaccine for the season , vaccination should ideally be offered during September or October. However , vaccination should continue throughout the season as long as influenza viruses are circulating.
- The Advisory Committee on Immunization Practices (The ACIP) recommends that all children 6 months old through 8 years old should get two doses of flu vaccine the first year that they are vaccinated the flu.
- Children 6 months through 8 years who require 2 doses should receive the first dose as soon as vaccine is available.
- Vaccination during July and August can be considered for children of any age who require only 1 dose.
- If the child had their first flu shot in a previous year but only got a single shot , then this year , they should get a flu shot and a booster shot.

Eligible groups for influenza vaccination :

➨ People who are :

- Are 65 years of age and over
- are a child aged 2 to 12 years
- are pregnant
- all those aged 6-23 months and 13-64 years at increased risk of influenza- related complications such as:

- Those with chronic illness ,e.g. chronic heart disease (including acute coronary syndrome) chronic liver disease , chronic neurological disease (where the neurological condition compromises clearance of respiratory secretions), chronic renal failure , chronic respiratory disease (including chronic obstructive pulmonary disease , cystic fibrosis , moderate or severe asthma , and bronchopulmonary dysplasia), diabetes mellitus , or haemoglobinopathies

- Those with immunosuppression due to disease or treatment , including asplenia or hyposplenism , and all cancer patients.
- Those with any condition that can compromise respiratory function (e.g. spinal cord injury , seizure disorder , or other neuromuscular disorder) especially those attending special schools / day centres
- Children and adults with Down syndrome
- Children with moderate to severe neurodevelopmental disorders such as cerebral palsy and intellectual disability
- Those with morbid obesity (Body mass index > 40)

- Residents of nursing homes , old people's homes , and other long stay facilities where rapid spread is likely to follow introduction of infection
- Healthcare workers
- Household contacts of people with underlying chronic health condition or Down syndrome
- Out-of-home care givers for people who have an underlying chronic health condition or have Down syndrome.
- People in regular contact with pigs , poultry or waterfowl

Interchangeability of influenza vaccines :

- Different age-appropriate brands (Influvac Tetra or Vaxigrip Tetra) are interchangeable for people who needs 2 doses of influenza vaccine in a single season.

Special considerations for concomitant use of PCV13 , DTaP and Influenza vaccines :

- The findings showed that **PCV13** and **DTaP** both increase the risk of **febrile seizure** when given **at the same time as the influenza vaccine.**

Flu Vaccine and People with Egg Allergies :

- Studies have clearly demonstrated that egg-allergic people can receive influenza vaccines. All egg –allergic individuals may be vaccinated against influenza using any of the vaccines authorized for use.(Canadian Immunization Guide)
- Every 6 months and older with egg allergy should receive an annual flu vaccine (egg based or non-egg based) that is otherwise appropriate for the recipient's age and health status can be used. (CDC)

Human Papillomavirus vaccine

Human papillomavirus Vaccination (minimum age : 9 years)

➨ **Routine and catch-up vaccination :**

- HPV vaccination routinely recommended **at age 11-12 years (can start at age 9 years)** and catch-up HPV vaccination recommended for all people through age 18 years if not adequately vaccinated.
- **2- or 3-dose series depending on age at initial vaccination :**

- **Age 9-14 at initial vaccination** : 2-dose series at 0, 6-12 months (minimum interval : 5 months ; repeat dose if administered too soon)
- **Age 15 years or older at initial vaccination** : 3-dose series at 0 , 1-2 months, 6 months (minimum intervals:dose1to dose 2: 4 weeks / dose 2 to 3 : 12 weeks / dose1 to dose 3 : 5 months; repeat dose if administered too soon)

- **Interrupted schedules :** If vaccination schedule is interrupted , the series does not need to be restarted.
- No additional dose recommended when any HPV vaccine series has been completed using the recommended dosing intervals

➨ **Special situations :**

- **Immunocompromising conditions, including HIV infection :**

3-dose series , even for those who initiate vaccination at age 9 through 14 years.

- **History of sexual abuse or assault : Start at age 9 years.**

- Two vaccines to prevent human papilloma (HPV) infection, the primary cause of cervical cancer , are now approved for use in many countries ; Cervarix and Gardasil.
- Some people Think of HPV immunization as only for girls but it's just as important for boys to get vaccinated.
- HPV vaccination protects boys from many kinds of cancer , most commonly from HPV-related cancers of the mouth and throat.
- The American Academy of Pediatrics recommends that HPV vaccination for healthy children can begin as early as age 9 years.

HPV VACCINE SCHEDULE

GARDASIL 9-14 years of age		GARDASIL 15 -45 years of age	
Dose 1	At first visit	Dose 1	At first visit
Dose 2	6-12 months after 1st dose	Dose 2	2 months after 1st dose
		Dose 3	6 months after 1st dose

Health Education & Research Organization (HERO)

HPV VACCINE SCHEDULE

CERVARIX 9-14 years of age		CERVARIX 15 years and older	
Dose 1	At first visit	Dose 1	At first visit
Dose 2	6 months after 1st dose	Dose 2	1 months after 1st dose
		Dose 3	6 months after 1st dose

Health Education & Research Organization (HERO)

What is the difference between GARDASIL and CERVARIX ? Are they interchangeable ?

- **CERVARIX** is a bivalent vaccine that protects against oncogenic(cancer causing) HPV types 16 and 18.
- **GARDASIL** vaccine is a quarivalent vaccine that protects against HPV types 16 and 18 as well as HPV types 6 and 11; the latter two are important causes of genital warts.
- **Whenever possible , one brand of vaccine should be used to complete a vaccine series.**
- If a person has received 1 or 2 HPV vaccine doses previously , and the brand of the previously received doses is not known , CERVARIX may be used to complete the series. Both vaccines provide protection against HPV types 16/18 and therefor clients are likely to achieve protective antibody levels against these HPV types. If less than 3 doses of GARDASIL are administered, counsel clients that protection against HPV types 6/11 (genital warts) cannot be assured.

Covid-19 vaccine

Routine vaccination (minimum age : 6 months [Moderna & Pfizer BioNTech COVID--19 vaccines], 12 years [Novavax COVID-19])

- Primary series :

- Age 6 months – 4 years : 2-dose series at 0,4-8 weeks (Moderna) or 3-dose series at 0, 3-8 , 11-16 weeks (Pfizer-BioNTech).
- Age 5-11 years : 2-dose series at 0,4-8 weeks (Moderna) or 2-dose series at 0, 3-8 weeks (Novavax, Pfizer – BioNTech).
- Age 12-18 years : 2-dose series at 0,4-8 weeks (Moderna) or 2-dose series at 0, 3-8 weeks (Novavax, Pfizer-BioNTech).

Special situations :

People who are moderately or severely immunocompromised

- Primary series

- Age 6 months – 4 years : 3-dose series at 0,4,8 weeks (Moderna) or 3-dose series at 0,3,11weeks (Pfizer-BioNTech).
- Age 5-11 years : 3-dose series at 0,4,8 weeks (Moderna) or 3-dose series at 0,3,7 weeks (Pfizer-BioNTech).
- Age 12-18 years : 3-dose series at 0,4,8 weeks (Moderna) or 2-dose series at 0,3 weeks (Novavax) or 3-dose series at 0,3,7 weeks (Pfizer-BioNTech).

Are COVID-19 vaccines interchangeable ?

Yes and no

- **Children 6 months through 4 years old** should generally **stick** to the same COVID vaccine brand.
- **But people ages 5 years and older** can make the choice to switch up which vaccine they receive. The FDA and CDC are allowing **a "mix-and-match" approach for COVID booster shots in this older age group**. This means that regardless of the vaccine that someone received first , they can choose the vaccine that they want next. **For example** , if someone initially received **the original Pfizer COVID vaccine** , they can choose which vaccine they want for their next dose. They could decide to get **Pfizer again , or they could choose to get Moderna or Novavax.**
- If a person originally received the Pfizer vaccine, switching to the Moderna vaccine has been shown to result in a good immune response. The same goes for switching to Pfizer if a person originally got Moderna.

Summary of recent changes (last updated October 6 , 2023) :

- The updated 2023-2024 formulation of Novavax COVID-19 Vaccine is recommended for people ages 12 years and older as follows :

- Initial vaccination : 2 doses of updated (2023-2024 formula) Novavax COVID-19 Vaccine.
- Previously vaccinated with any Original monovalent or bivalent COVID-19 (Moderna,Novavax, Pfizer-BioNTech, Janssen): 1 dose of updated (2023-2024 Formula) Novavax Vaccine.

- People who are moderately or severely immunocompromised may receive 1 or more additional updated (2023-2024 Formula) Novavax vaccine doses.
- People ages 12 years and older have the option of receiving either the updated (2023-2024 Formula) mRNA (Moderna,Pfizer-BioNTech) or updated (2023-2024 Formula) Novavax vaccine.

Thank You ☺